
CPSC 483 Project Proposal

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Abstract

Social media has transformed society. Information can be spread and consumed at faster rates than ever before. This has allowed misinformation to spread at faster rates than ever before, so detecting fake news or misinformation when it is posted is a crucial problem to solve. There have been many advances in this problem, first by applying the field of machine learning to it in general, and then recently by applying graph neural networks to analyze the complicated relationships and plentiful information that can be used in assisting with fake news detection. Cutting-edge models use features like temporal information or construct heterogeneous graphs, and we propose two approaches that build upon temporal information usage and heterogeneous graph usage. Namely, these are the Temporal Graph Attention Network (TGAT) and the Heterogeneous Graph Transformer (HGT).

1 Problem Definition

The internet, especially social media, has transformed society by offering an option to share and consume content extremely fast and conveniently. This results in many positive impacts, but also poses challenges due to the lack of regulation and verification on many social media platforms. This has enabled a rampant spread of misinformation, which poses a significant threat to public discourse, political stability, public health, and even the economy [9]. The challenge of distinguishing the truth from intentional or unintentional misinformation on platforms such as social media is difficult. The ability to identify and detect fake news quickly and accurately is crucial to mitigate the harmful potential effects of misinformation and to prevent its proliferation, especially during crucial times such as pandemics or elections.

Traditionally, machine learning methods for fake news detection have relied on the text content of articles, focusing on sentiment analysis and linguistic features [1]. However, fake news detection is a complex phenomenon, and it can involve many things including content and social interactions (retweets, comments, profiles, etc.). The complexity of this detection and the potential information involved has resulted in graphs being an ideal model for these structures. Thus, there is a growing shift toward applying Graph Neural Networks (GNNs) to model these relationships and capture the structural dependencies within social networks [5]. GNN-based approaches have shown strong and promising results by leveraging knowledge-driven, propagation-based, and heterogeneous social context-based methods.

Many earlier GNN-based models employ only homogeneous graphs [8, 6, 2], treating all nodes and relationships equally. This fails to capture the potential nuance of interactions within social networks. Some modern models incorporate more features such as temporal information and heterogeneous graph structures. Models that consider temporal features include DHGNN [16] and TGNF [15] which use static temporal snapshots, and models that consider heterogeneous graphs structures include GLAN [17] and FANG [3].

We propose two approaches to this problem that build upon these directions: a **Temporal Graph Attention Network (TGAT)** and a **Heterogeneous Graph Transformer (HGT)**. We discuss preliminary approaches in section 4.

The objective of developing these new methods is primarily to improve the accuracy of fake news detection. However, since we are trying to add additional context to the graph structure and information, there is also potential for increased interpretability that we will evaluate throughout the project. Since we are examining and trying to implement more complicated models, efficiency will not be a priority.

We also want to note that TGAT [13] and HGT [7] are already existent graph structures. Although we are currently keeping the names, the goal is to either (1) show that they are effective when implemented in a new domain (fake news detection in particular domains) or (2) improve upon the architectures using novel changes or combining them with current models for fake news detection and seeing if there is an improvement over current models.

2 Datasets

We consider the most commonly used fake news detection datasets, including Twitter15 [10], Twitter16 [10], PHEME-5 [18], PHEME-9 [12], Weibo [11], and FakeNewsNet [14].

Of these datasets, the only one with data from the past 5 years is FakeNewsNet, so we choose to use that dataset. Additionally, we thought it would be interesting to test our models on a separate dataset. Knowing that misinformation plays a huge role in public health crises, we chose the COVID-19 Healthcare Misinformation Dataset (CoAID) [4].

FakeNewsNet is a dataset composed mostly of two topics: politics and entertainment. A lot of social context information is included, such as news, news sources, news spreaders, retweets, reply propagation graphs, etc. It is constructed using data from social media and news sources, using news labeled by the fact-checking platforms PolitiFact and GossipCop. FakeNewsNet includes thousands of news articles and millions of related social media interactions, providing rich information for both content and propagation-based analysis.

CoAID (Covid-19 heAlthcare mIsinformation Dataset) is a dataset that contains diverse COVID-19 healthcare misinformation. This includes fake news on websites and social media platforms, users' social engagement about such news, and more. CoAID includes 4,251 news articles, 296,000 related user engagements, 926 social media posts about COVID-19, and ground truth labels. CoAID will hopefully provide temporal data that allows the modeling of misinformation across one big event. We show statistics of CoAID in table 1.

	Fake		True	
	Claim	News Article	Claim	News Article
# Information on Website	28	204	454	3,565
# Tweets	484	10,439	8,092	141,652
# Replies	626	7,436	12,451	114,820
# Social Platform Posts	650 (880)	-	42 (46)	-
# Total	1,788 (2,018)	18,079	21,039 (21,043)	260,037

Table 1: Statistics of CoAID Version 0.3; adapted from [4].

Combined, these two datasets provide comprehensive data about misinformation in three different topics: politics, entertainment, and public health. Based on potential changes as the project progresses, we may choose to use the more classic, larger datasets listed above.

3 Related Works

Information for this section is largely based off of information from Gong et al.'s 2023 survey of Fake News Detection through Graph-based Neural Networks [5].

3.1 Temporal Features

Previous approaches that utilize temporal features primarily use static temporal snapshots. We will highlight two of them, Dynamic Heterogeneous Graph Neural Network (DHGNN) [16] and Temporally Evolving Graph Neural Network for Fake News Detection (TGNF) [15].

Dynamic GNN is a variant of Graph Neural Networks that captures the evolving structure of graphs by taking static temporal snapshots of the propagation graph at different time points. These graph snapshots are encoded by a GNN to generate graph representations, which are then processed using sequence modeling approaches.

TGNF considers temporal information at the graph node level. It utilizes a temporal pattern-aware GNN named Temporal Graph Network (TGN) to encode the evolution of propagation graphs. TGNF utilizes TGN to build a graph structure based on the sequence of user interactions (e.g., replies and retweets on social media). The graph is encoded by a Graph Convolutional Network. Then, it is fed into a fully-connected neural network to predict whether or not the news is fake.

3.2 Heterogeneous Graphs

Another advantage of social media data is that it often contains information beyond semantic and temporal data. In particular, social media posts can be related through structures like likes, reposts, and tagging, which all provide additional graph-like data.

The **global-local attention network (GLAN)** jointly encodes local semantic information with global structural data. Local semantic information is better represented by integrating data from multiple related posts using the attention mechanism. Global relationships between original posts and reposts, as well as the authors of those posts, are all integrated for a richer relationship network.

Another source of additional information is **learnt context during an event**. Specifically, Conditional Random Fields can be leveraged when a high quantity of data is available surrounding a specific event or topic. Instead of querying specific pieces of information for rumor detection—as many other models do—the classifier can draw upon previous data on the event to identify rumors.

4 Proposed Approaches

Our research question is as follows. How can we incorporate features such as temporal information or graph heterogeneity into modern GNN techniques for fake news detection to increase detection accuracy and interpretability?

We believe that there is promise in further modeling graph heterogeneity, as well as examining the temporal relationships between nodes and edges on the graph and hypothesize that by further testing in these directions, we can find improvements in accuracy as measured by AUC.

4.1 Temporal Graph Attention Network (Andrew)

A Temporal Graph Attention Network (TGAT) aims to capture evolving user interactions as well as changes in propagation patterns through temporal attention mechanisms. The attention mechanism will focus on the most influential interactions over time, making it possible to distinguish between levels of influence and critical moments over time. This can build upon previous work using temporal information by not restricting to static time snapshots.

Nodes in the graph will represent entities like users and articles. Edges will represent their relationships, such as shares and comments. Each edge and node will also have a timestamp to detail when the interaction occurred. We will likely use BERT text embeddings as a node feature as well to get a semantic representation of the news.

We will encode time into the nodes through either sinusoidal encodings or learned positional encodings, similar to how positions are embedded in transformers. We will conduct tests to see which is more effective and efficient. Finally, similar to Graph Attention Networks, we will then use a temporal attention layer to aggregate messages from neighboring nodes. The difference is that TGAT needs to consider temporal relevance – we can do this by modifying the attention score based on temporal differences between interactions.

For propagation, since we are dealing with a large graph, we can also only sample a subset of neighbors based on temporal information. Then, we can aggregate information scores from this subset of neighbors.

We will determine the final model architecture as well as loss function (likely cross-entropy) and potential regularization through ablation testing.

4.2 Heterogeneous Graph Transformer (Annli)

HGT will handle the heterogeneous nature of social media data by modeling different types of nodes (users, articles, comments) and relationships (shares, replies, follows) in a unified architecture, representing the heterogeneous nature of social media more effectively.

Similar to Graph Attention Networks (GAT), we will use attention to compute the importance of different neighbors. However, with a heterogeneous graph, we will need to adapt this mechanism. Currently, we think the best approach will be to allow each node type to have a different projection matrix so that it can learn a different transformation for each node type. Then, the attention score between neighbors will differ based on their types. This makes sense because it classifies interactions between articles and users differently.

In the aggregation step, we can use type-specific aggregators and type-specific linear transformations. We can still initialize each node with features such as textual embeddings from BERT.

Additional features, such as activation functions and specific network architecture will be optimized through additional research and ablation testing.

5 Evaluation Metrics

To evaluate the performance of our models, we will tell each model, including baselines, to classify each piece of news in a dataset as real or fake. Then, we will compare the accuracy of each model using the AUC score.

Based on whether or not we keep the datasets or have to modify the models that we are creating, the evaluation metrics may change. However, we are primarily focused on whether or not a new model is able to better utilize supporting information to differentiate between fake and real articles, so accuracy will be the main target. This is how we will primarily evaluate our hypothesis that our novel methods will lead to improvement in fake news detection.

We will also attempt to evaluate precision, recall, and F1 scores, as was done when evaluating different methods in the FakeNewsNet dataset.

6 Timeline

We (Andrew and Annli) will work independently for background information, collaborate when brainstorming each other's models, create specific models ourselves, and share a central repository for baselines and the specific tests. We will implement models independently but combine results and do analysis together. We will determine a good way to split the project report after we have a very solid idea of our results and methods. The timeline for the completion of the project is as follows:

- 10/13/24 - 10/26/24: Read and understand papers and code related to TGAT (Andrew) and HGT (Annli), and get familiar with training and testing models on the datasets used. Also finalize the model architectures that we would like to individually implement. Go to office hours to confirm our project progress.
- 10/27/24 - 11/9/24: Create the code for TGAT (Andrew) and HGT (Annli) and test on datasets, comparing to current baselines. If initial results don't turn out well, brainstorm new model architectures and run tests again. Go to office hours to confirm our project progress.
- 11/10/24 - 11/16/24: Compare results, conduct further testing, including ablation testing, and talk about what we could do to further enhance our project report. Go to office hours to confirm our project progress.
- 11/17/24 - 12/5/24: Write project report.

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